EEES 302 Digital Design

Experiment 6

Flips Flops

Objective of Experiment:

The objective of this laboratory is to introduce the use of flip-flops.

Review of Experiment:

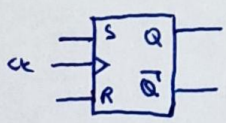
A multivibrator is a circuit with an output that oscillates between logic high and low states.

The primary focus of this experiment is the Bistable multivibrators, or flip-flops, which has two stable states ($Q=0$, $Q=1$).

Flip-Flops:

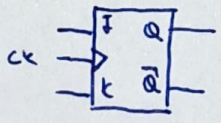
A flip-flop is the simplest type of memory cell. Unlike combinational logic, a flip-flops output depends on its current inputs and the sequence in which they were applied.

SR flip-flop



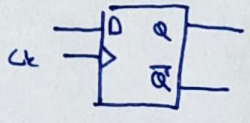
S	R	Q(t+1)
0	0	Q
0	1	0
1	0	1
1	1	illegal

JK Flip-Flop



J	K	Q(t+1)
0	0	Q(t)
0	1	0
1	0	1
1	1	$\overline{Q(t)}$

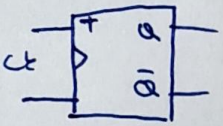
D Flip-Flop



D	Q(t+1)
0	0
1	1

$Q(t+1) = D$

T Flip-Flop



T	Q(t+1)
0	Q(t)
1	$\overline{Q(t)}$

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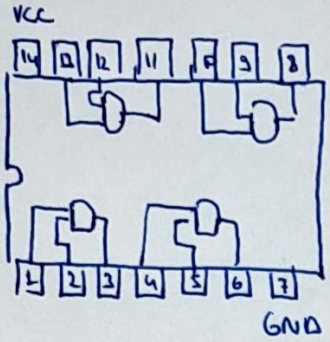
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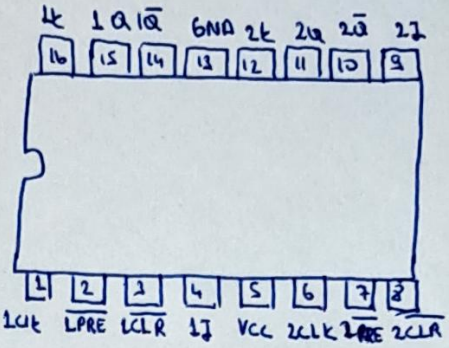
Flip Flops

Equipment List:

74LS08 AND gate IC



7476 JK Flip-Flop IC

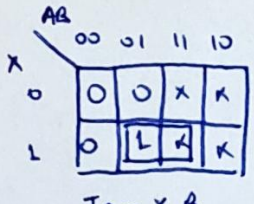


Jumpers and Breadboard is used to built circuit.

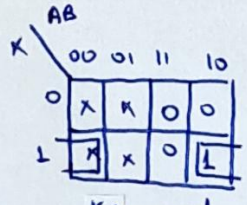
Ossilator and multimeter is used for measurements.

Preliminary Work:

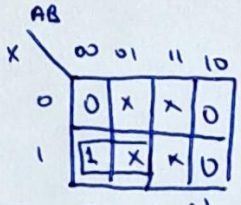
	P.S		N.S					
X	A	B	A	B	J_A	K_A	J_B	K_B
0	0	0	0	0	0	X	0	X
0	0	1	0	1	0	X	X	0
0	1	0	1	0	X	0	0	X
0	1	1	1	1	X	0	X	0
1	0	0	0	1	0	X	1	X
1	0	1	1	0	1	X	X	1
1	1	0	1	1	X	0	X	0
1	1	1	0	0	X	1	0	X



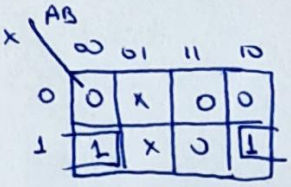
$J_A = X.B$



$K_A = X.B'$



$J_B = X.A'$



$K_B = X.A$

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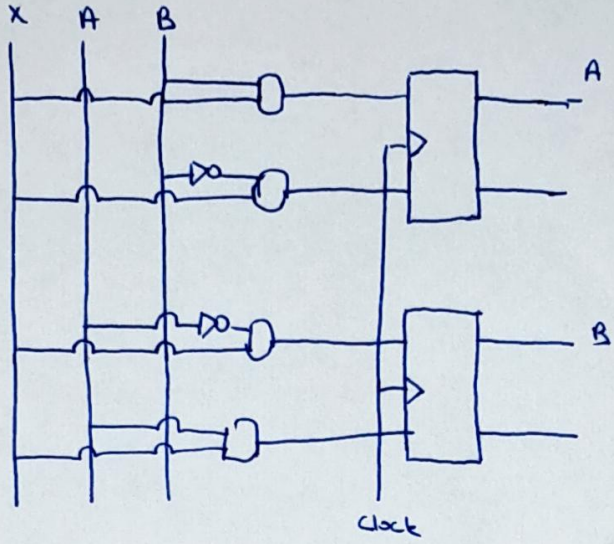
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Circuit Implamentation:



Experimental Procedure:

- Connected VCC (5V) and GND from the power supply.
- Clock is given from clock generator and connected to clock pin.
- The input is connected to logic high 5V and logic low 0V to test.

The truth table is tested by changing the input X. Output is measured from ossilator and the pulse signals is observed.

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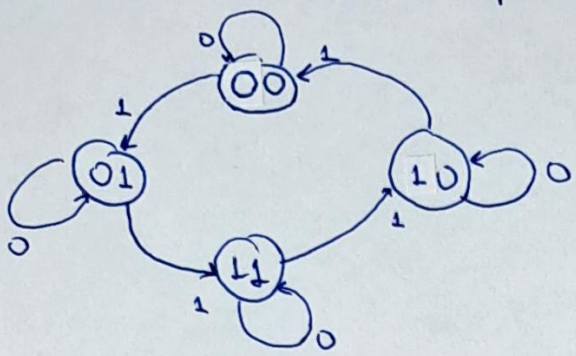
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Result:

The circuit is given the state digram order when the input is 1 and, it hold the state when the input is 0.



We observed a repeating square wave that has 1/4th the frequency of our clock signal.

We designed a working synchronous state machine using Jk flip-flops as foundational memory cells.
Additionally, we learned to implement debouncing circuits to prevent mechanical switches from causing false clock triggers.

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